

extra credit

1.

rigid tank, well insulated  
 $v = \text{constant}$   $Q = 0$

mixture of ammonia  
 saturated gas =  $0.005 \text{ m}^3$   
 saturated fluid =  $1.5 \text{ m}^3$   
 $P_1 = 5 \text{ bar}$

(a) at saturated vapor state  
 while  $P = 5 \text{ bar}$   $v_g = 1.5800 \times 10^{-3} \frac{\text{m}^3}{\text{kg}}$

$$m_g = \frac{V}{v} = \frac{0.005}{1.5800 \times 10^{-3}} = m_g = 8.62 \text{ kg}$$

$g = 0.2503$

$$m_f = \frac{V}{v} = \frac{1.5}{0.2503} = 5.99 \text{ kg}$$

$$m = m_g + m_f = 8.62 + 5.99 = 14.61 \text{ kg}$$

$$x = \frac{m_g}{m} = \frac{5.99}{14.61} = 0.41$$

$$v_{\text{mix}} = v_f + x(v_g - v_f)$$

$$= (1.5800 \times 10^{-3}) + 0.41(0.2503 - 1.5800 \times 10^{-3})$$

$$v_{\text{mix}} = 0.1036 \frac{\text{m}^3}{\text{kg}}$$

$$u_f = u_f + x(u_g - u_f) \quad \text{at pressure } 5 \text{ bar}$$

$$= 198.39 + 0.41(1321.02 - 198.39)$$

$$u_1 = 658.67 \text{ kJ/kg}$$

$$u_g = 198.39$$

$$u_f = 1321.02$$

for state 2

$$v_1 = v_2 = 0.1036 \frac{\text{m}^3}{\text{kg}}$$

| Bar                 | $\frac{\text{m}^3}{\text{kg}}$ | $\frac{\text{kJ}}{\text{kg}}$ |
|---------------------|--------------------------------|-------------------------------|
| $P_2 = 14$          | $v_2 = 0.0923$                 | $u_2 = 1339.56$               |
| $P_1 = 12$          | $v_1 = 0.1075$                 | $u_1 = 1337.52$               |
| $P_x = \text{find}$ | $v_x = 0.1036$                 | $u_x = \text{find}$           |

interpolation

$$\frac{P_2 - P_1}{P_2 - P_x} = \frac{v_2 - v_1}{v_2 - v_x}$$

$$P_2 - P_x = \frac{(P_2 - P_1)(v_2 - v_x)}{(v_2 - v_1)} = \frac{(14 - 12)(0.0923 - 0.1036)}{(0.0923 - 0.1075)}$$

$$14 - P_x = 1.497$$

$$14 - 1.497 = P_x = 12.51 \text{ bar}$$

$$\text{Pressure}_2 = 12.51 \text{ bar}$$

$$\Delta U = u_2 - u_1$$

$$= (1338.04 - 658.67) \frac{\text{kJ}}{\text{kg}} \times (14.61 \text{ kg})$$

$$\Delta U = 9925.6 \text{ kJ}$$

(2) 18 kg of ammonia, initially at 100 kPa and  $-10^{\circ}\text{C}$

Ammonia compress to a pressure of 550 kPa.

The pressure and specific volume are related by  $PV^{1.1} = \text{constant}$ .

(a) determine the work.

$$P_1 = 100 \text{ kPa} \quad P_2 = 550 \text{ kPa}$$

$$T_1 = -10^{\circ}\text{C}$$

$$n = 1.1$$

$$\Delta U = Q - W$$

state 1 at  $P_1 = 100 \text{ kPa}$   $\rightarrow v_1 = 1.5834 \text{ m}^3/\text{kg}$

$T_1 = -10^{\circ}\text{C}$   $u_1 = 1325.85 \text{ kJ/kg}$

$$P_1 v_1^{1.1} = P_2 v_2^{1.1}$$

$$\left(\frac{100 \text{ kPa}}{550 \text{ kPa}}\right)^{1/1.1} = \left(\frac{v_1}{v_2}\right)^{1.1}$$

$$(0.1818)(1.5834)^{1.1} = v_2^{1.1} = 0.3014 \frac{1}{1.1}$$

$$v_2 = (0.3014)^{1/1.1} = 0.3361 \text{ m}^3/\text{kg}$$

at  $P_2 = 550 \text{ kPa}$ ,  $v_2 = v_1 = 0.3361 \text{ m}^3/\text{kg}$

$$v_2 = 0.34190$$

$$v_1 = 0.32307$$

$$v_x = 0.3361$$

$$u_2 = 1537.02$$

$$u_1 = 1520.36$$

$$u_x = \text{find}$$

$$\frac{v_2 - v_1}{v_2 - v_x} = \frac{u_2 - u_1}{u_2 - u_x}$$

$$u_2 - u_x = \frac{(v_2 - v_x)(u_2 - u_1)}{(v_2 - v_1)}$$

$$1537.02 - u_x = \frac{(0.34190 - 0.32307)(1537.02 - 1520.36)}{(0.34190 - 0.32307)}$$

$$1537.02 - 11.29 = u_x = 1525.73 \text{ kJ/kg}$$

(a)  $\text{work} = \frac{P_2 v_2 - P_1 v_1}{1 - n}$

$$= \frac{(550)(0.3361) - (100)(1.5834)}{1 - 1.1}$$

$$= -265.15 \frac{\text{kJ}}{\text{kg}} \times 18 \text{ kg}$$

$$W = -4772.7 \text{ kJ}$$

(b) determine heat transfer

$$\Delta U + \Delta KE + \Delta PE = Q + W$$

$$Q = \Delta U + W$$

$$= (u_2 - u_1) + W$$

$$= (1525.73 - 1325.85) \frac{\text{kJ}}{\text{kg}} \times 18 \text{ kg} + (-4772.7 \text{ kJ})$$

$$= (3597.85 - 4772.7) \text{ kJ}$$

$$Q = -1174.85 \text{ kJ}$$



③ A Piston-cylinder

$m_{air} = 0.25 \text{ kg}$ , initially at  $P_1 = 2 \text{ bar}$   
 $T_1 = 200^\circ\text{C}$

air is heated at constant pressure

$$V_2 = 2V_1$$

Assume ideal gas model

Constant Specific heat ratio  $\gamma = 1.4$

(a) determine initial volume

$$PV = mRT$$

$$V = \frac{mRT}{P}$$

$$V = \frac{(0.25)(0.287)(473) \text{ kg} \cdot \frac{\text{kJ}}{\text{kg} \cdot \text{K}}}{200 \text{ kPa}}$$

$$\left\{ \begin{array}{l} R = 0.287 \frac{\text{kJ}}{\text{kg} \cdot \text{K}} \\ T = 200^\circ\text{C} + 273 \frac{\text{K}}{^\circ\text{C}} = 473^\circ\text{K} \\ m = 0.25 \text{ kg} \\ P = 2 \text{ bar} \times 10^5 \text{ Pa} = 200 \text{ kPa} \end{array} \right.$$

$$V_1 = 0.1697 \text{ m}^3$$

(b) work = ?

$P = \text{constant} = 2 \text{ bar}$

$$V_1 = 0.1697 \text{ m}^3 \quad V_2 = 2(0.1697) = 0.3394 \text{ m}^3$$

$$W = -\int P dV = -(2 \times 10^5)(0.3394 - 0.1697)$$

$$= -33940 \text{ J} = -33.94 \text{ kJ}$$

(c) Heat transfer Q = ?  $\Delta KE + \Delta PE + \Delta U = Q - W$

for air, while  $\gamma = 1.4$

$$C_p = 1.003$$

$$Q = m \times C_p (T_2 - T_1)$$

$$= 0.25 \text{ kg} \times 1.003 \frac{\text{kJ}}{\text{kg} \cdot \text{K}} (473 - 200)^\circ\text{K}$$

$$Q = 118.6 \text{ kJ}$$

from ideal gas law

$$\frac{V_1}{V_2} = \frac{T_1}{T_2}$$

$$T_2 = \frac{T_1 V_2}{V_1} = \frac{473 \text{ K} (0.3394 \text{ m}^3)}{0.1697 \text{ m}^3} = 946 \text{ K}$$

\*information collected from wikipedia

④ Air is compressed

$$P_1 = 0.7 \text{ bar} \quad T_1 = 275 \text{ K} \quad V_1 = 0.25 \text{ m}^3$$

to final,  $V_2 = 0.03 \text{ m}^3$ , Process described by  $PV^{1.25} = \text{constant}$ .

Assume ideal gas behavior, neglect  $KE^0, PE^0$   
using constant specific heats evaluated at  $T_1$ .

(a) Determine work done, in kJ

$$P_1 V_1^{1.25} = P_2 V_2^{1.25}$$

$$P_1 = 0.7 \text{ bar}$$

$$V_1 = 0.25 \text{ m}^3$$

$$V_2 = 0.03 \text{ m}^3$$

$$\left(\frac{V_1}{V_2}\right)^{1.25} P_1 = P_2$$

$$\left(\frac{0.25}{0.03}\right)^{1.25} (0.7) = P_2 = 9.91 \text{ bar}$$

$$\text{work} = \frac{P_2 V_2 - P_1 V_1}{1-n} = \frac{(9.91 \times 10^5 \text{ Pa})(0.03) - (0.7 \times 10^5 \text{ Pa})(0.25)}{(1-1.25)} = -48.92 \text{ kJ}$$

(b) Determine heat transfer

$$Q = \Delta U + W$$

$$\Delta U = U_2 - U_1$$

$$Q = c \times m \times \Delta T$$

$$m = \frac{V}{v}$$

$c = \text{Specific heat capacity of the system}$